

04_Random_Forest_Classification

EP

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Random Forest : Classification

For comparison with current literature, below I run the richness and Simpson random forest models using classification over regression. The richness and Simpson diversity indices are split based on median, creating two diversity labels: “high” and “low”. This mirrors the approach taken by recent studies in which two groups of samples are matched, e.g. 500 “flood” and 500 “no flood” samples.

```
library(targets)
library(tidymodels)
library(ranger)
library(sf)
library(dplyr)
```

```
set.seed(399)
```

Load Data

Read in the final grid model and transform the simpson scores. This isn't strictly necessary for Random Forest but allows for direct comparison with the beta regression model.

```
setwd("~/CSCT_Pipeline")

grid_model_all <- tar_read(grid_model_all)

df <- grid_model_all %>%
  sf::st_drop_geometry()

n <- nrow(df)
df$simpson_adj <- (df$simpson * (n - 1) + 0.5) / n

predictors <- c("n_days", "dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
  "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
  "dist_to_river", "pct_greenpace")
```

Model Setup : Richness and Simpson

```

#Classification setup
df <- df %>%
  mutate(
    richness_class = factor(if_else(sp_richness >=
      median(sp_richness, na.rm = TRUE),
      "high", "low"),
      levels = c("low", "high")),
    simpson_class = factor(if_else(simpson_adj >=
      median(simpson_adj, na.rm = TRUE),
      "high", "low"),
      levels = c("low", "high")))

#RF Specification
rf_spec_class <- rand_forest(
  mtry = tune(),
  min_n = tune(),
  trees = 500) %>%
  set_engine("ranger", importance = "permutation",
    respect.unordered.factors = "order", probability = TRUE) %>%
  set_mode("classification")

folds <- group_vfold_cv(df, group = city)

fold_city_lookup <- folds %>%
  mutate(city = purrr::map_chr(
    splits, ~ as.character(unique(assessment(.x)$city)))) %>%
  select(id, city)

#RF Recipe and Workflow
richness_recipe_class <- recipe(
  as.formula(paste("richness_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

richness_wf_class <- workflow() %>%
  add_recipe(richness_recipe_class) %>%
  add_model(rf_spec_class)

simpson_recipe_class <- recipe(
  as.formula(paste("simpson_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

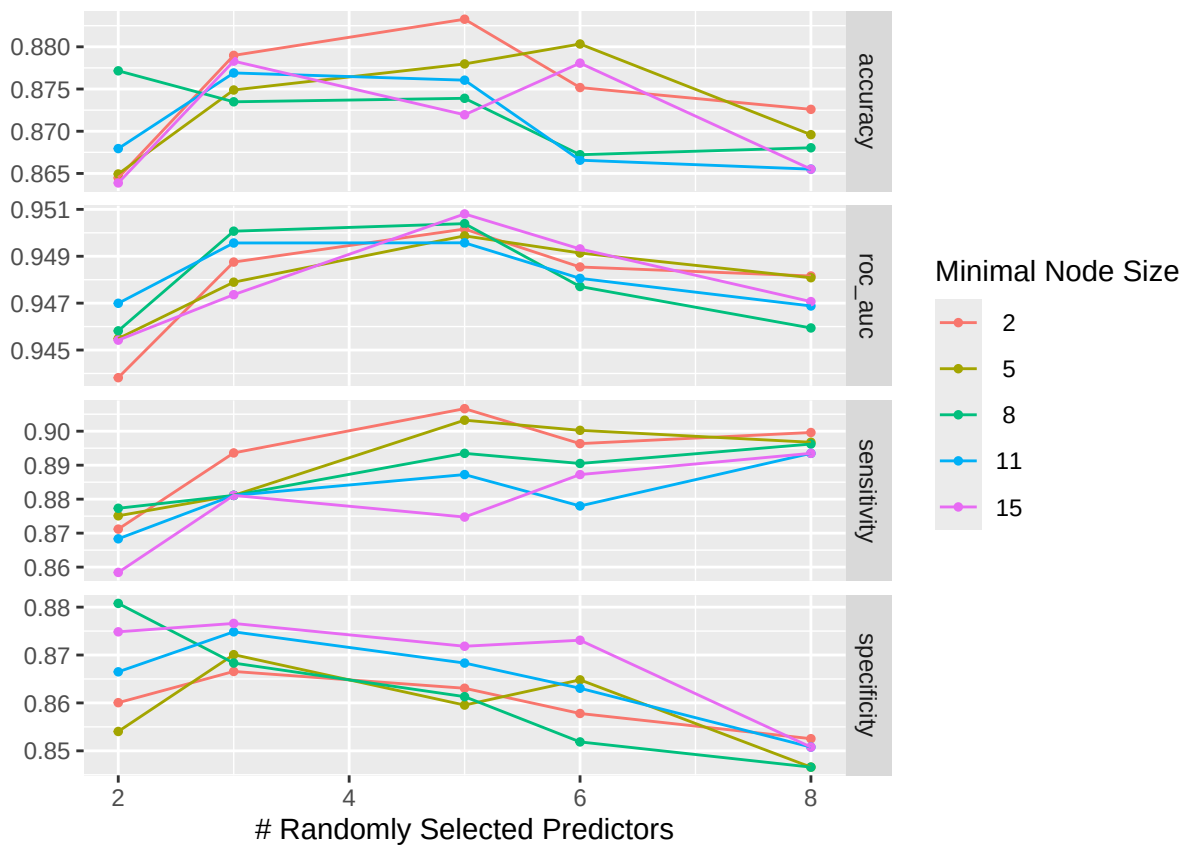
simpson_wf_class <- workflow() %>%
  add_recipe(simpson_recipe_class) %>%
  add_model(rf_spec_class)

rf_grid <- grid_regular(
  mtry(range = c(2, 8)),
  min_n(range = c(2, 15)),
  levels = 5) #Total of 25 combinations in the grid.

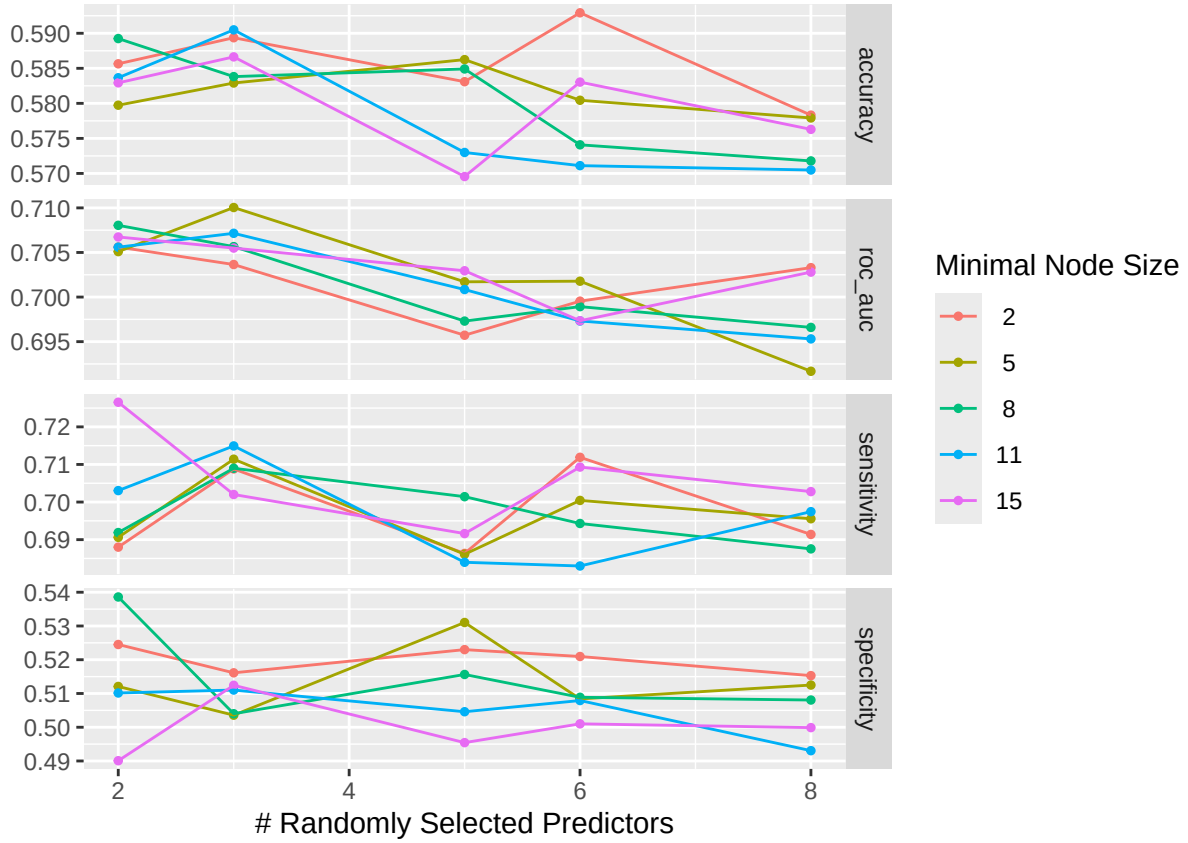
```

Hyperparameter Tuning

```
richness_res_class <- tune_grid(  
  richness_wf_class,  
  resamples = folds,  
  grid      = rf_grid,  
  metrics   = metric_set(roc_auc, accuracy, sensitivity, specificity))  
  
simpson_res_class <- tune_grid(  
  simpson_wf_class,  
  resamples = folds,  
  grid      = rf_grid,  
  metrics   = metric_set(roc_auc, accuracy, sensitivity, specificity))  
  
autoplot(richness_res_class)
```



```
autoplot(simpson_res_class)
```



Select Best Hyperparameters

```
show_best(richness_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
  rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
  knitr::kable(caption = "Top 5 hyperparameter combinations:
    species richness class (ranked by ROC AUC)")
```

Table 1: Top 5 hyperparameter combinations: species richness class (ranked by ROC AUC)

mtry	min_n	Mean ROC AUC	Std. error
5	15	0.951	0.010
5	8	0.950	0.011
5	2	0.950	0.011
3	8	0.950	0.013
5	5	0.950	0.012

```
show_best(simpson_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
```

```

rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
knitr::kable(caption = "Top 5 hyperparameter combinations:
Simpson diversity class (ranked by ROC AUC)")

```

Table 2: Top 5 hyperparameter combinations: Simpson diversity class (ranked by ROC AUC)

mtry	min_n	Mean ROC AUC	Std. error
3	5	0.710	0.022
2	8	0.708	0.020
3	11	0.707	0.022
2	15	0.707	0.017
3	8	0.706	0.023

```

best_richness_class <- select_best(richness_res_class, metric = "roc_auc")
best_simpson_class <- select_best(simpson_res_class, metric = "roc_auc")

```

The richness model has a best ROC AUC of 0.951, and performance is stable across the grid. This suggests that the model is not highly sensitive to tuning. However, the simpson model is noticeably weaker (ROC AUC = 0.710) and more variability between tuning models. This suggests that the richness model classifies better than the simpson diversity using these predictor variables.

Prediction Rate AUC

This looks at the mean AUC across held-out city folds, and individual fold's AUC scores.

```

prediction_rate_richness <- collect_metrics(richness_res_class) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc")

prediction_rate_simpson <- collect_metrics(simpson_res_class) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

prediction_rate_richness_byfold <- collect_metrics(
  richness_res_class, summarize = FALSE) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc")

prediction_rate_simpson_byfold <- collect_metrics(
  simpson_res_class, summarize = FALSE) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

overall_richness <- collect_metrics(richness_res_class) %>%
  filter(mtry == best_richness_class$mtry,

```

```

      min_n == best_richness_class$min_n,
      .metric == "roc_auc") %>%
pull(mean)

overall_simpson <- collect_metrics(simpson_res_class) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc") %>%
pull(mean)

best_richness_class <- select_best(richness_res_class, metric = "roc_auc")

best_simpson_class <- select_best(simpson_res_class, metric = "roc_auc")

overall_richness <- collect_metrics(richness_res_class) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc") %>%
pull(mean)

overall_simpson <- collect_metrics(simpson_res_class) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc") %>%
pull(mean)

city_richness <- collect_metrics(richness_res_class, summarize = FALSE) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc") %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(richness_auc = .estimate)

city_simpson <- collect_metrics(simpson_res_class, summarize = FALSE) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc") %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(simpson_auc = .estimate)

city_auc <- city_richness %>%full_join(city_simpson, by = "city")

overall_auc <- tibble(city = "Overall",
                     richness_auc = overall_richness,
                     simpson_auc = overall_simpson)

classification_auc_table <- bind_rows(overall_auc, city_auc) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename("City (Held-Out)" = city,
         "Richness ROC AUC" = richness_auc,
         "Simpson ROC AUC" = simpson_auc)

```

```
classification_auc_table %>%
  knitr::kable(
    caption = "Overall and city-level cross-validated ROC AUC for species
richness and Simpson diversity")
```

Table 3: Overall and city-level cross-validated ROC AUC for species richness and Simpson diversity

City (Held-Out)	Richness ROC AUC	Simpson ROC AUC
Overall	0.951	0.710
Nottingham	0.918	0.700
Exeter	0.936	0.775
Newcastle	0.957	0.639
Bristol	0.971	0.705
Gateshead	0.972	0.731

```
# Check for any folds where AUC couldn't be computed (single-class fold)
richness_na_folds <- prediction_rate_richness_byfold %>%
  filter(is.na(.estimate))

simpson_na_folds <- prediction_rate_simpson_byfold %>%
  filter(is.na(.estimate))

richness_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

```
simpson_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

As seen before, there is more variability in the Simpson scores compared to the richness scores when looking at individual cities. This suggests that the richness model extrapolates to other cities better than the simpson model.

Final Model

This is fit on the entire dataset.

```
final_richness_wf_class <- finalize_workflow(
  richness_wf_class, best_richness_class)
richness_fit_class <- fit(final_richness_wf_class, data = df)

final_simpson_wf_class <- finalize_workflow(
  simpson_wf_class, best_simpson_class)
simpson_fit_class <- fit(final_simpson_wf_class, data = df)
```

Success Rate AUC

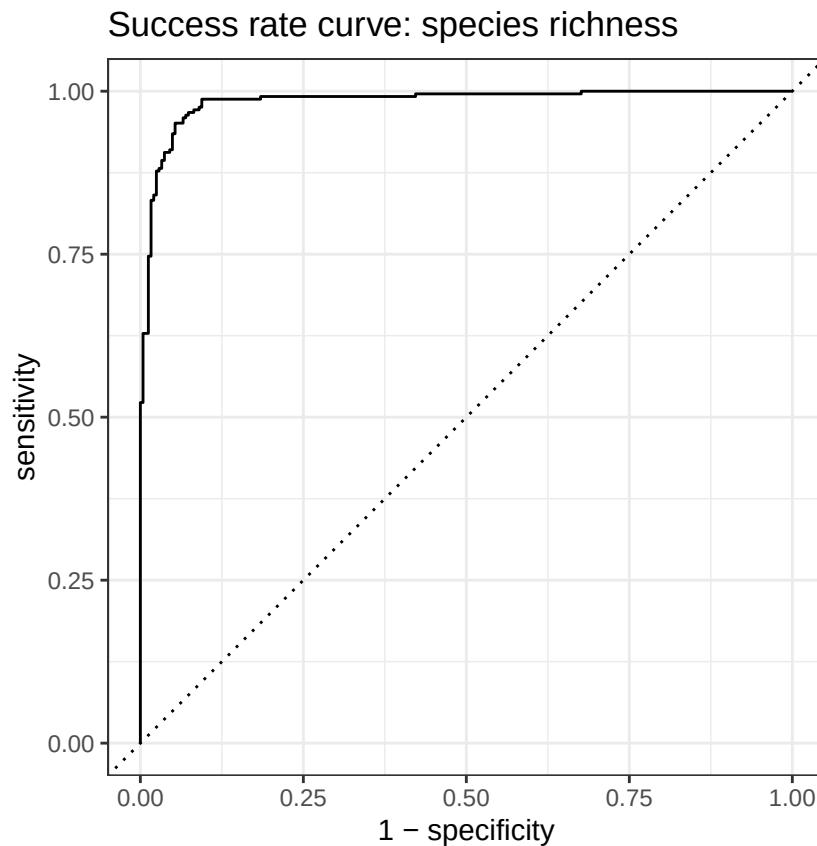
```
richness_success <- augment(richness_fit_class, new_data = df)
simpson_success  <- augment(simpson_fit_class,  new_data = df)

success_rate_richness <- roc_auc(richness_success, truth = richness_class,
                                .pred_high, event_level = "second")

success_rate_simpson <- roc_auc(simpson_success, truth = simpson_class,
                                .pred_high, event_level = "second")
```

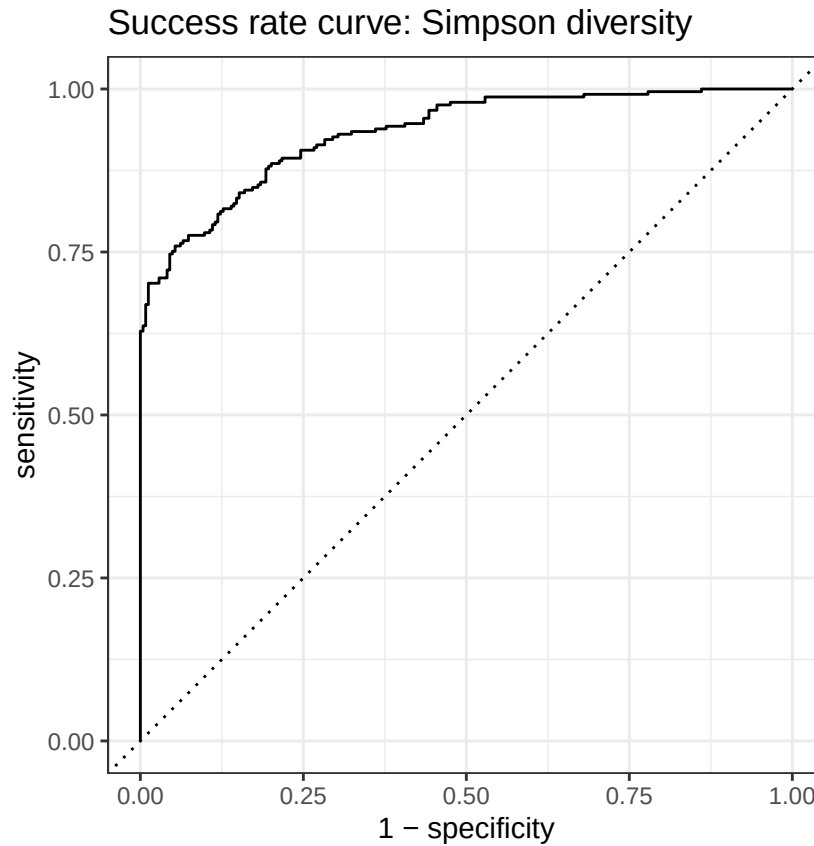
ROC Curves

```
roc_curve(richness_success,
          truth = richness_class,
          .pred_high,
          event_level = "second") %>%
  autoplot() + labs(title = "Success rate curve: species richness")
```



```
roc_curve(simpson_success,
          truth = simpson_class,
          .pred_high,
```

```
event_level = "second") %>%
autoplot() + labs(title = "Success rate curve: Simpson diversity")
```



Summary Table : Success Rate vs Prediction Rate AUC

```
tibble::tibble(
  Outcome = c("Species richness", "Simpson diversity"),
  `Success rate AUC (in-sample)` = c(success_rate_richness$.estimate,
                                       success_rate_simpson$.estimate),
  `Prediction rate AUC (cross-city)` = c(prediction_rate_richness$mean,
                                           prediction_rate_simpson$mean),
  `Prediction rate SE` = c(prediction_rate_richness$std_err,
                           prediction_rate_simpson$std_err)
) %>%
mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
knitr::kable(caption = "Random forest classification performance:
                    success rate vs. prediction rate AUC")
```

Table 4: Random forest classification performance: success rate vs. prediction rate AUC

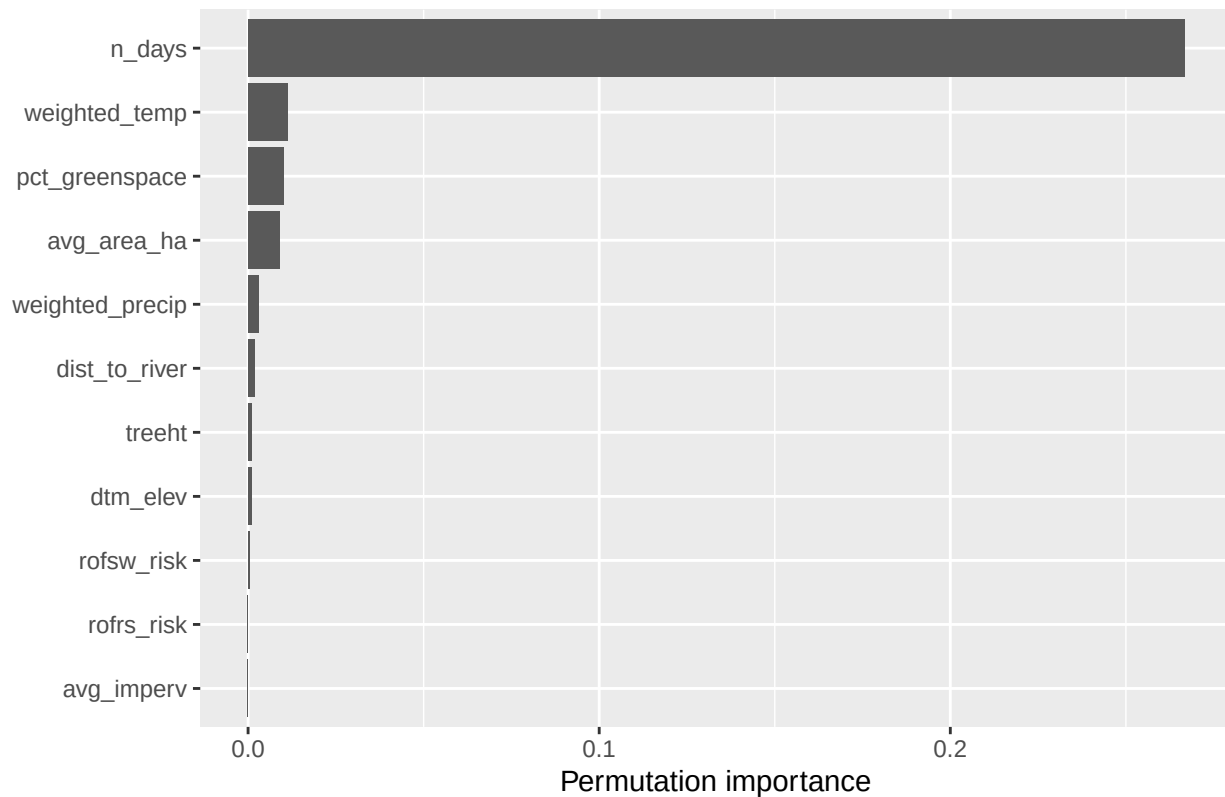
Outcome	Success rate AUC (in-sample)	Prediction rate AUC (cross-city)	Prediction rate SE
Species richness	0.984	0.951	0.010
Simpson diversity	0.933	0.710	0.022

For the richness model, there is a small gap between the sample AUC and cross-city AUC (0.984 to 0.951) suggesting that the model is capturing real patterns, not sure memorizing city specific patterns. On the other hand, the Simpson model has a larger gap (0.933 to 0.710). This suggests that the model fits the training data well, but struggles to generalise to other, unseen, cities. This is backed up by the previous, city level breakdown of AUC scores where there was more variability in the Simpson scores.

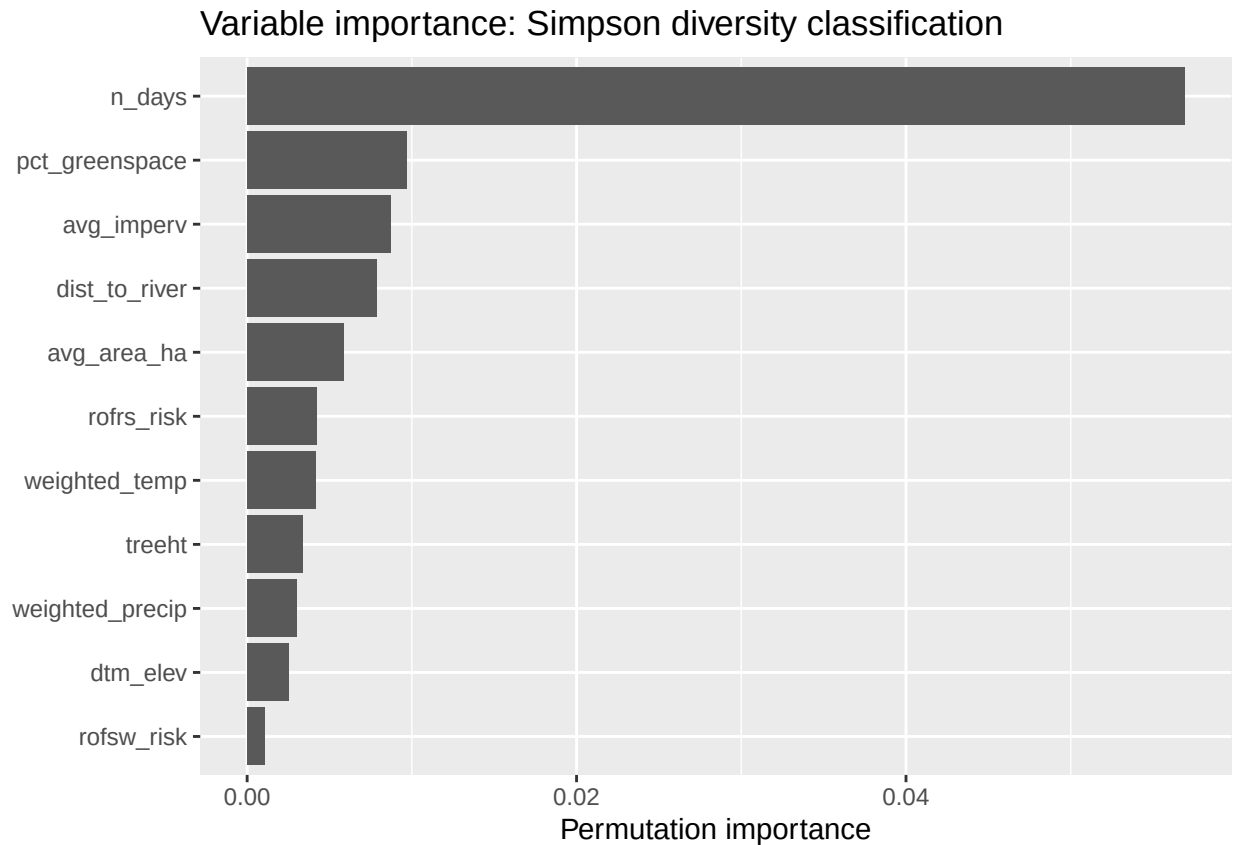
Variable Importance

```
richness_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
  as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col() +
  coord_flip() +
  labs(title = "Variable importance: species richness classification",
       x = NULL, y = "Permutation importance")
```

Variable importance: species richness classification



```
simpson_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
  as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col() +
  coord_flip() +
  labs(title = "Variable importance: Simpson diversity classification",
        x = NULL, y = "Permutation importance")
```



Once again, **n_days** is the dominant predictor variable for both models, likely representing sampling effort over true biodiversity. In the richness model, **weighted_temp**, **pct_greenpace**, and **avg_area_ha** show secondary importance. **weighted_temp** and **pct_greenpace** were also relevant in the regression random forest and significant in the negative binomial regression. In the Simpson model, **pct_greenpace**, **avg_imperv**, **dist_to_river**, and **avg_area_ha** show secondary importance.

To conclude, the richness classifier model is strong and demonstrates transferability between cities. There is evidence that the predictors capture real ecological signal. Whereas the Simpson model is weaker with the training data fitting well but it struggles to generalise to new cities. To attempt to look closer at the true ecological drivers, as opposed to sampling effort, these models are rerun in the next notebook without the predictor **n_days**.